

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
Course Coordinator Name		Venkataramana Veeramsetty	
Instructor(s) Name		Dr. V. Venkataramana (Co-ordinator)	
		Dr. T. Sampath Kumar	
		Dr. Pramoda Patro	
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		Dr.J.Ravichander	
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		Dr. Anirodh Kumar	
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		Mr. Kundhan Kumar	
		Ms. Ch.Rajitha	
		Mr. M Prakash	
		Mr. B.Raju	
		Intern 1 (Dharma teja)	
		Intern 2 (Sai Prasad)	
		Intern 3 (Sowmya)	
		NS_2 (Mounika)	
Course Code	24CS002PC215	Course Title	AI Assisted Coding
Year/Sem	II/I	Regulation	R24
Date and Day of Assignment	Week11 - Monday	Time(s)	
Duration	2 Hours	Applicable to Batches	
AssignmentNumber:21.1(Present assignment number)/24(Total number of assignments)			
Q.No.	Question		Expected Time to complete
1	Lab 21 – Collaborative AI Coding: Pair Programming and Version Control Integration Lab Objectives: - Learn to use AI as a pair programmer for writing, reviewing, and optimizing code. - Understand how to integrate AI-generated code into version		Week11 - Monday

	control systems (Git). Practice collaborative workflows such as commit messages, branching, and merging.	
	Task 1 – AI-Assisted Pair Programming - Use AI to generate an initial program (e.g., a simple To-Do List Manager). - Ask AI to suggest enhancements (such as error handling, additional features, or refactoring). - Compare the original and improved versions to understand the role of AI in collaboration.	
	Task 2 – Version Control Integration - Initialize a Git repository for your project. - Perform the following steps: - Create and switch to a new branch. - Add AI-assisted modifications. - Commit with a meaningful message. - Merge changes back into the main branch. - Reflect on how version control supports collaborative development with AI assistance.	
	Task 3 – AI for Commit Message Generation - Use AI to generate concise, meaningful commit messages based on the changes made. - Compare AI-generated messages with human-written ones. - Discuss how AI can improve documentation and traceability in Git history.	
	Task 4 – Pair Programming Simulation - Student A writes the initial version of a program. - Student B, with AI assistance, suggests improvements or adds features. - Both students collaborate using Git branching and merging to integrate their work. ✓ Deliverables (For All Tasks) - AI-generated prompts for code and test case generation. - At least 3 assert test cases for each task. - AI-generated initial code and execution screenshots. - Analysis of whether code passes all tests.	

	5. Improved final version with inline comments and explanation. 6. Compiled report (Word/PDF) with prompts, test cases, assertions, code, and output.	
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